

8.6

a. $N=100$, $SSE_M = 97161.9174$

$$\hat{\sigma}_F^2 = 12.024^2$$

$$\hat{\sigma}_M^2 = \frac{97161.9174}{577-4} = 169.567$$

$$\textcircled{1} \begin{cases} H_0 : \sigma_M^2 = \sigma_F^2 \\ H_a : \sigma_M^2 \neq \sigma_F^2 \end{cases}$$

$$\textcircled{2} \alpha = 0.025$$

$$\textcircled{3} F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} \stackrel{H_0}{\sim} F(577-4, 423-4)$$

$$\textcircled{4} RR = \{F \geq F_{0.025}(419, 573) = 1.198\}$$

$$\textcircled{5} F_0 = \frac{169.567}{12.024^2} \approx 1.1724$$

$\textcircled{6} \because F_0 \notin RR$, don't reject H_0 .

b.

$$\textcircled{1} \begin{cases} H_0 : \sigma_M^2 = \sigma_S^2 \\ H_a : \sigma_M^2 > \sigma_S^2 \end{cases}$$

$$\textcircled{2} \alpha = 0.05$$

$$\textcircled{3} F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_S^2} \stackrel{H_0}{\sim} F(600-5, 400-5)$$

$$\textcircled{4} RR = \{F \geq F_{0.05}(595, 395) = 1.1647\}$$

$$\textcircled{5} F_0 = \frac{100,710,471 / 595}{36,231.082 / 395} = 1.1887$$

$\textcircled{6} F_0 \in RR$, reject H_0

C. ① $\begin{cases} H_0: \alpha_1 = \alpha_2 = \dots = \alpha_5 = 0 \\ H_a: \text{不全為 } 0 \end{cases}$

② $\alpha = 1.05$

③ $\chi^2 = NR^2 \sim \chi^2(5-1)$

④ $RR = \{ \chi^2 > \chi^2_{0.05}(4) = 9.488 \}$

⑤ $NR^2 = 59.03$

⑥ $NR^2 \in RR$, reject H_0

存在異質 Var 支持 (b), wage 變異可能隨其他特徵變動, 但無法確定不同婚姻下有差異。

D. $df = 4 + 2 + 6 = 12$

交互項 = $\frac{4 \times 3}{2} = 6$

④ $RR = \{ \chi^2 > \chi^2_{0.05}(12) = 21.03 \}$

⑤ $LM_0 = 98.82$

⑥ $LM_0 \in RR$, reject H_0 , 存在異質變異故

e.		se		robse	
	intercept	2.56	<	3.5	) wider
	EDUC	0.14	<	0.16	
	EXPER	0.031	>	0.029	) narrower
	MEIRO	1.05	>	0.84	
	FEMALE	0.61	>	0.8	

不存在 *inconsistent*, 存在異質變異性下, OLS 的 SE 有偏誤, 可能低估/高估真實 SE, Robse 可修正此偏差, 係數 SE $\uparrow/\downarrow$ 為正常

f. compatible.

(a): married 的 t 值為 1.0, 不顯著, married 對工資平均水準不影響。

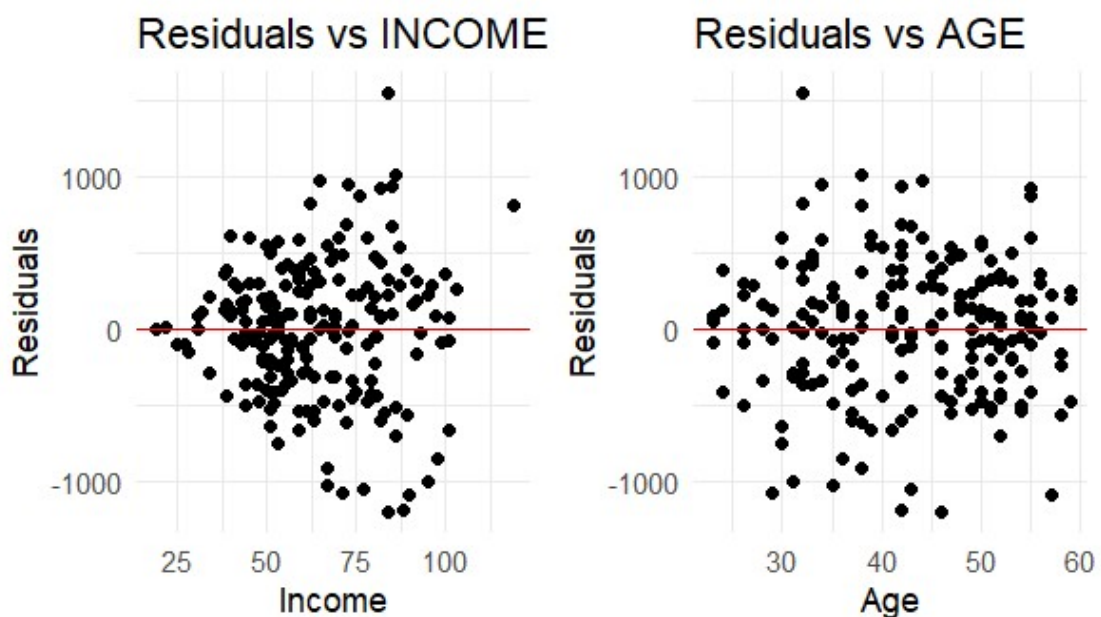
(b): married 會影響工資變異性

16(a) OLS Confidence Interval

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                        Lower_2.5 Upper_97.5
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KIDS 95% CI (OLS) -135.330   -28.323
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在控制收入與年齡不變的前提下，每增加一名子女對行駛里程數的影響，其 95% 信賴區間落於 -135.33 至 -28.323 英里之間

16(b)



觀察 Residuals vs INCOME 圖表可以發現，誤差項的分布不均，呈現隨收入上升而發散的趨勢，符合異質變異的特徵

16(c) Goldfeld-Quandt Test

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                        value
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GQ Statistic          3.104
df1 (numerator)        86
df2 (denominator)      86
p-value                0.00000
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根據 Goldfeld-Quandt (GQ) 檢定 結果，其 p 值為 0.00000，遠低於 0.05 的顯著水準。因此，我們拒絕虛無假設，並得出模型存在異質變異性的結論，意即

誤差變異量會隨收入增加而擴大。

16(d) Robust Confidence Interval

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	Lower_2.5	Upper_97.5

KIDS 95% CI (Robust)	-139.323	-24.330

16(e) GLS Confidence Intervals

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	Lower_2.5	Upper_97.5

KIDS CI (GLS Trad)	-119.894	-33.718
KIDS CI (GLS Robust)	-121.413	-32.199
